1. Overview

This report describes a small study of question answering over PDF documents. The study was run by the Riverbend Data Lab in March 2025. The team evaluated three retrieval methods on 120 questions. All experiments ran on a laptop with 16 GB of memory and no GPU.

2. Method

Documents were split into passages of 1000 characters with an overlap of 150 characters. Each passage was embedded with a MiniLM model that produces vectors of 384 dimensions. Keyword search used the BM25 algorithm. The two rankings were merged with reciprocal rank fusion using a constant of 60. The system returns at most five passages for each question.

3. Results

Hybrid search reached a recall of 91 percent at five passages. Keyword search alone reached 78 percent and embedding search alone reached 85 percent. The median answer time was 4.2 seconds. The slowest question took 19 seconds. Scanned pages were the main source of errors, causing 14 of the 31 failures.

4. Limitations

The study used only English and Hindi documents. Tables were converted to plain text, which sometimes lost row boundaries. The authors recommend repeating the study with at least 500 questions. No private documents were used; every file was public.

5. Data Collection

The documents came from three public sources: government reports, university course notes and open textbooks. In total the team collected 48 files with 1350 pages. Files larger than 50 megabytes were excluded. Each file was checked by two reviewers before it was added.

6. Question Writing

Two reviewers wrote the 120 questions by hand. Ninety questions had an answer in the documents and thirty did not. Each answerable question lists the page that holds the evidence. Questions were written in English, Hindi and a mixture of both.

7. Answer Generation

A local language model with four billion parameters wrote the answers. The model was told to say not found when the passages did not contain the answer. The prompt allowed at most 512 tokens of output. A cloud model was never used.

8. Conclusions

Combining keyword and embedding search gave the best recall without extra cost. The system refused to answer 27 of the 30 unanswerable questions. The authors plan to test a multilingual embedding model next. The code and the questions will be released under an open licence.